

Birzeit University

Discrete Mathematics (COMP 233): Quiz #1

Section 9: Answers & Marking Scheme

Second Semester 2025/2026 · Coverage: Chapters 1, 2, 3

Question One [45 pts]: Logical equivalence, tautology, and an inference rule

Q1.a [15 pts]

Simplify the following logical expression to an equivalent expression:

$$\neg(p \wedge \neg q) \wedge (p \vee q)$$

SOLUTION Apply equivalence laws step by step. The target equivalent expression is q .

#	Form	Law used
0	$\neg(p \wedge \neg q) \wedge (p \vee q)$	<i>given</i>
1	$\equiv (\neg p \vee \neg(\neg q)) \wedge (p \vee q)$	<i>De Morgan's law on $\neg(p \wedge \neg q)$</i>
2	$\equiv (\neg p \vee q) \wedge (p \vee q)$	<i>double negation: $\neg(\neg q) \equiv q$</i>
3	$\equiv q \vee (\neg p \wedge p)$	<i>distributive law (factor out q)</i>
4	$\equiv q \vee c$	<i>negation law: $\neg p \wedge p \equiv c$</i>
5	$\equiv q$	<i>identity law: $q \vee c \equiv q$</i>

FINAL $\neg(p \wedge \neg q) \wedge (p \vee q) \equiv q$.

Marking, Q1.a [15 pts, 3 pts per correct step]

Pts	Criterion
3	Step 1: De Morgan applied correctly to $\neg(p \wedge \neg q)$.
3	Step 2: double negation applied to $\neg(\neg q)$.
3	Step 3: distributive law to factor out q .
3	Step 4: negation law identifying $\neg p \wedge p \equiv c$.
3	Step 5: identity law reaching q .

NOTES Accept equivalent orderings of the laws. Accept *NOT*, $\sim$, or $\neg$ in place of $\neg$. If the student names a different law that yields the same intermediate form correctly, award full marks for that step.

Q1.b [25 pts]

Show that the following expression is a **tautology** by proving the logical equivalence using a **truth table**:

$$(p \wedge (p \rightarrow q)) \rightarrow q \equiv T$$

SOLUTION Build the truth table column by column. Four rows for two variables p, q :

p	q	$p \rightarrow q$	$p \wedge (p \rightarrow q)$	$(p \wedge (p \rightarrow q)) \rightarrow q$
T	T	T	T	T
T	F	F	F	T
F	T	T	F	T
F	F	T	F	T

The final column is **T in every row**, so the expression evaluates to true for every assignment of truth values to p and q .

VERDICT $(p \wedge (p \rightarrow q)) \rightarrow q \equiv T$. The expression is a **tautology**.

Marking, Q1.b [25 pts]

Pts	Criterion
3	Correct column headers (the five columns above).
3	Correctly computed intermediate columns: $p \rightarrow q$ and $p \wedge (p \rightarrow q)$.
16	4 pts per correct row (4 rows). Lose 1 pt per wrong cell.
3	Correct conclusion stating the expression is a tautology and equivalent to T .

NOTES Award 0 pts if the truth table is missing entirely. If only the final column is given without intermediate columns, cap at 8 pts.

Q1.c [5 pts]

Which Rule of Inference is represented by the tautology in part (b)? Write its **formal argument form**.

RULE **Modus Ponens.** The tautology $(p \wedge (p \rightarrow q)) \rightarrow q$ says: *if both p and $p \rightarrow q$ are true, then q is true.*

FORM

$$\begin{array}{l} p \\ p \rightarrow q \\ \hline \therefore q \end{array}$$

Marking, Q1.c [5 pts]

Pts	Criterion
2	Correctly naming Modus Ponens .
3	Correct formal argument form: two premises (p and $p \rightarrow q$) and conclusion $\therefore q$. All three lines must be present.

NOTES Accept equivalent notations for the conclusion ($\therefore q$, $\vdash q$, "therefore q"). 0 pts if the rule name is wrong.

Question Two [55 pts]: Conditional variations, predicates & quantifiers, short answers

Q2.a [20 pts]

Write the **contrapositive**, **converse**, **inverse**, and **negation** of the following statement:

"If Sara passes Discrete Math, then she will register for Data Structures and she will take Algorithms."

SETUP Let p = "Sara passes Discrete Math", q = "Sara registers for Data Structures", r = "Sara takes Algorithms". The original statement is $p \rightarrow (q \wedge r)$.

CONTRAPOSITIVE

$$\neg(q \wedge r) \rightarrow \neg p \equiv (\neg q \vee \neg r) \rightarrow \neg p$$

"If Sara does not register for Data Structures or does not take Algorithms, then she does not pass Discrete Math."

CONVERSE

$$(q \wedge r) \rightarrow p$$

"If Sara registers for Data Structures and takes Algorithms, then she passes Discrete Math."

INVERSE

$$\neg p \rightarrow \neg(q \wedge r) \equiv \neg p \rightarrow (\neg q \vee \neg r)$$

"If Sara does not pass Discrete Math, then she does not register for Data Structures or does not take Algorithms."

NEGATION

$$\neg(p \rightarrow (q \wedge r)) \equiv p \wedge (\neg q \vee \neg r)$$

"Sara passes Discrete Math, and she does not register for Data Structures or does not take Algorithms."

IMPORTANT The negation is *not* a conditional. It is a conjunction: the hypothesis holds and the conclusion fails.

Marking, Q2.a [20 pts, 5 pts per variation]

Pts	Criterion (apply to each of the four variations)
3	Correct symbolic form for the variation.
2	Correct verbal sentence in English.

NOTES De Morgan must be applied to $\neg(q \wedge r)$ in the contrapositive, inverse, and negation. **Deduct 2 pts** overall if missing throughout. Accept $\neg q \vee \neg r$ or $\neg r \vee \neg q$. **0 pts** for the negation if it is written as a conditional.

Q2.b [20 pts]

Predicates on the domain $D = \{2, 3, 5, 6, 8\}$:

$E(x)$ = "x is even"; $P(x)$ = "x is prime".

For each of (i)–(ii), write the statement formally with quantifiers and predicates, then give its truth value with justification. For the Rewrite, give the English sentence and the truth value with justification.

(i) Every element in D is even. **[5 pts]**

(ii) There exists an element in D that is both even and prime. **[5 pts]**

Rewrite the following formal statement as a complete English sentence and give its truth value with justification: **[10 pts]**

$$\forall x \in D, (E(x) \rightarrow P(x))$$

(I) Formal: $\forall x \in D, E(x)$. Truth value: **False**. Justification: $x = 3 \in D$ is not even. (Also $x = 5$.)

(II) Formal: $\exists x \in D, (E(x) \wedge P(x))$. Truth value: **True**. Justification: $x = 2 \in D$ is even and prime.

REWRITE English: "For every element x in D , if x is even then x is prime."

Truth value: **False**. Justification: $x = 6 \in D$ is even ($6 = 2 \cdot 3$) but is not prime; the hypothesis holds and the conclusion fails. ($x = 8$ is another counterexample.)

Marking, Q2.b [20 pts]

Pts	Criterion
2	(i) Correct formal statement $\forall x \in D, E(x)$.
1	(i) Correct truth value (False).
2	(i) Justification with a specific counterexample ($x = 3$ or $x = 5$).
2	(ii) Correct formal statement $\exists x \in D, (E(x) \wedge P(x))$.
1	(ii) Correct truth value (True).
2	(ii) Justification with a specific witness ($x = 2$).
4	Rewrite: correct English sentence (universal conditional read aloud).
2	Rewrite: correct truth value (False).
4	Rewrite: justification with a specific counterexample where the hypothesis holds but the conclusion fails ($x = 6$ or $x = 8$).

NOTES A bare truth value with no specific witness/counterexample loses the justification marks. Generic claims ("there is one") without a named element score 0 for that step.

Q2.c [15 pts, 3 pts each]

Provide the direct logical equivalent or operation for each of the following.

- (1) Write the **negation** of $p \rightarrow q$.
- (2) Which variation of $p \rightarrow q$ is logically equivalent to its **Converse**?
- (3) Given $p \leftrightarrow q$, write it as a conjunction of two conditional statements.
- (4) Write the **formal argument form** of Modus Tollens (using p and q).
- (5) Let $D = \{1, 2, 3, 4\}$ and let $P(x) = "x \text{ is a factor of } 8."$ Write the **truth set** of $P(x)$.

(1) $\neg(p \rightarrow q) \equiv p \wedge \neg q$. Reading: " p is true and q is false." The negation of a conditional is not itself a conditional.

(2) **Inverse:** $\neg p \rightarrow \neg q \equiv q \rightarrow p$. The converse and inverse are contrapositives of each other and are therefore logically equivalent.

(3) $p \leftrightarrow q \equiv (p \rightarrow q) \wedge (q \rightarrow p)$.

(4) Modus Tollens:

$$\begin{array}{l} p \rightarrow q \\ \neg q \\ \hline \therefore \neg p \end{array}$$

Reading: "If p then q . Not q . Therefore not p ."

(5) Truth set: $\{1, 2, 4\}$. The factors of 8 are 1, 2, 4, 8; intersect with $D = \{1, 2, 3, 4\}$. 8 is a factor of 8 but $8 \notin D$, so it is excluded.

Marking, Q2.c [15 pts, 3 pts each]

Pts	Criterion
3	(1) Correct answer $p \wedge \neg q$. Accept $\neg(p \rightarrow q)$ written as-is but only 1 pt if not simplified.
3	(2) Correctly naming the Inverse . 1 pt if only the symbolic form $\neg p \rightarrow \neg q$ is given without naming it.
3	(3) Correct answer $(p \rightarrow q) \wedge (q \rightarrow p)$. Accept equivalent ordering. 0 pts if disjunction or wrong connective is used.
3	(4) Modus Tollens: 1 pt for premise $p \rightarrow q$, 1 pt for premise $\neg q$, 1 pt for conclusion $\therefore \neg p$. All three lines must appear for full marks.
3	(5) Correct truth set $\{1, 2, 4\}$. Deduct 1 pt if 8 is included (domain restriction missed). Deduct 1 pt per other wrong or missing element.

Marks summary

Question One	15 + 25 + 5	45
Question Two	20 + 20 + 15	55
Total		100

General Marking Principles

- For proof-style and equivalence-step questions, marking is sequential: students lose individual marks for missing or incorrect steps, not the whole question.
- Accept either negation glyph throughout: $\neg$, $\sim$, or $\neg$.
- In Q2.a, the negation must not be expressed as a conditional. De Morgan must be applied to $\neg(q \wedge r)$ in the contrapositive, inverse, and negation.
- In Q2.b, justifications without a specific witness or counterexample score 0 for the justification step.
- In Q2.c, item 4 (Modus Tollens) requires all three lines (two premises and the conclusion) for full marks.
- If a student names a different law in Q1.a that yields the same intermediate form correctly, award full marks for that step.